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დურიშვილი ნიკა

გველესიანი თენგიზ

სენიორ პროექტი

მექანიკური ულტრაბგერითი სკანერი

ხელმძღვანელი:

ვარდიაშვილი გუგა

სულაბერიძე ზვიად

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ანოტაცია

ტრადიციული დიაგნოსტიკური და არადესტრუქციული ულტრაბგერითი სისტემები დიდწილად ეყრდნობა მრავალელემენტიან Phased Array ტექნოლოგიას. მიუხედავად იმისა, რომ Phased Array გვთავაზობს დინამიურ მართვას და მაღალ კადრების სიხშირეს, მათთან დაკავშირებული მრავალარხიანი სხივის ფორმირების ელექტრონიკა და მაღალი წარმოების ხარჯები მათ უკიდურესად ძვირადღირებულს ხდის ლაბორატორიული კურსებისა და შეზღუდული რესურსებით კვლევისთვის. ეს პროექტი ნერგავს ავტომატიზირებული, დაბალფასიანი ულტრაბგერითი სკანირების პლატფორმას, რომელიც ცვლის რთულ ელექტრონულ სხივურ მართვას ერთელემენტიანი მექანიკური პოზიციონირებით.

პრორქტის ძირითადი მიზანი იყო ჩაძირული სამიზნეების საიმედო B-სკანირების და ფსევდო-3D მოცულობითი ვიზუალიზაციის მიღწევა ერთი 5 MHz აკუსტიკური სენსორის გამოყენებით. ამის მისაღწევად, ავაწყეთ ელექტრომექანიკური სისტემა და სიგნალების დამუშავების სქემა. შევქმენით PCB, რომელიც უშვებს დაახლოებით 100ვ მაღალი ძაბვის იმპულსებს, რომლებიც საჭიროა transducer-ის მართვისთვის. იმავე PCB-ში ვკითხულობთ სუსტ დაბრუნებულ აკუსტიკურ ექოებს, ვაძლიერებთ, ვფილტრავთ და ვადისკრეტიზირებთ. Digilent Arty S7 FPGA ამ მაღალსიჩქარიან სიგნალებს ამუშავებს. FPGA კოორდინაციას უწევს იმპულსებს, ექოს მონაცემებს აგროვებს და გადასცემს ციფრულ აკუსტიკურ ჩანაწერებს კომპიუტერს UART-ის საშუალებით.

სკანირების კონტეინერზე მექანიკურად გადაადგილდება gantry plate, რომელიც დამზადებულია 3D პრინტერით და სტანდარტული მეტალის ნაწილების გამოყენებით. X ღერძის გასწვრივ მოძრაობას მართავს ერთი ბიპოლარული stepper ძრავა, ხოლო y ღერძი იყენებს ორ სინქრონიზებულ stepper ძრავას, რათა თავიდან ავიცილოთ მექანიკური დაჭიმვა გადაღების პროცესში. Java-ში დაწერილი დესკტოპის აპლიკაცია ამუშავებს შემომავალ მონაცემთა ნაკადს, ითვლის სიგნალის მაგნიტუდებს, ასახავს სივრცულ კოორდინატებს და ასახავს რეალურ დროში B-სკანირების ნაწილებს.

ჩაძირული სამიზნეების გამოყენებით ექსპერიმენტულმა ვალიდაციამ აჩვენა, რომ პლატფორმა საიმედოდ კითხულობს აკუსტიკურ ანარეკლებს, განასხვავებს სტრუქტურულ საზღვრებს და ხედავს სხეულის შიდა გეომეტრიას. შედეგად მიღებული პროტოტიპი ადასტურებს, რომ FPGA-ზე დაფუძნებული ულტრაბგერითი სკანერის შერწყმა მექანიკურ სისტემასთან ეფექტურ, დაბალფასიან ალტერნატივას გვთავაზობს არადესტრუქციული შეფასებისთვის.

Abstract

Conventional diagnostic and non-destructive ultrasound imaging systems rely heavily on multi-element phased array transducers. While phased arrays offer dynamic beam steering and high frame rates, their associated multichannel analog front-ends, complex beamforming electronics and high manufacturing costs make them prohibitively expensive for laboratory coursework and resource-constrained research. This project presents the design and implementation of an automated, low-cost ultrasound scanning platform that replaces complex electronic beam steering with a single-element mechanical positioning approach.

The primary engineering objective was to achieve reliable cross-sectional B-scan and pseudo-3D volumetric imaging of submerged targets using a single 5 MHz acoustic transducer. To support this, a complete electromechanical and signal-processing pipeline was constructed. A custom analog front-end (AFE) printed circuit board was designed to generate the approximately 100 V high-voltage excitation pulses required to drive the transducer and to condition weak returning acoustic echoes through low-noise amplification and filtering. A Digilent Arty S7 FPGA handles high-speed signal processing and timing control. The FPGA coordinates pulse excitation, manages echo data acquisition and streams digitised acoustic records to a host personal computer over a serial communication link.

Mechanical displacement across the scanning tank is handled by a custom Cartesian gantry fabricated using fused filament 3D-printed structural components and standard metal hardware. Motion along the x-axis is driven by a single bipolar stepper motor, while the y-axis employs two synchronized stepper motors to maintain carriage alignment and prevent mechanical racking during raster patterns. A desktop application developed in Java processes the incoming data stream, computes signal envelopes, maps spatial coordinates and renders real-time B-scan slices alongside reconstructed 2.5D representations of the scanned volume.

Experimental validation using submerged phantom targets demonstrated that the platform reliably captures acoustic reflections, distinguishes structural boundaries and resolves internal geometry. Mechanical positional repeatability was sufficient to prevent distortion across successive acquisition lines without requiring closed-loop optical feedback. The resulting prototype confirms that combining FPGA-based acquisition with single-transducer

mechanical translation offers an effective, low-cost alternative for basic non-destructive evaluation and practical ultrasound training.

Acknowledgments

We would like to express our sincere gratitude to our senior project supervisors, Guga Vardiashvili and Zviad Sulaberidze, for their continuous technical guidance and practical advice throughout the electronic design, debugging and experimental evaluation of this ultrasound scanning system. We also thank our colleague Anzor Revazishvili who helped us solder very little parts on the PCB.

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Figures, Tables and Abbreviations

This section presents all system tables, component specifications, hardware schematics, PCB drawings, testbench photographs, oscilloscope captures and signal processing diagrams for the ultrasound scanner platform, formatted according to IEEE standards.

List of Abbreviations and Acronyms

- **ADC:** Analog-to-Digital Converter
- **AFE:** Analog Front End
- **BOM:** Bill of Materials
- **CAD:** Computer-Aided Design
- **DAC:** Digital-to-Analog Converter
- **DC:** Direct Current
- **DSP:** Digital Signal Processing
- **FIR:** Finite Impulse Response
- **FPGA:** Field Programmable Gate Array
- **GUI:** Graphical User Interface
- **HV:** High Voltage
- **I/Q:** In-Phase and Quadrature
- **LDO:** Low Dropout Regulator
- **LNA:** Low Noise Amplifier
- **LPF:** Low Pass Filter
- **MSPS:** Mega-Samples Per Second
- **PCB:** Printed Circuit Board
- **PLA:** Polylactic Acid
- **PMOD:** Peripheral Module Interface
- **PRF:** Pulse Repetition Frequency
- **RF:** Radio Frequency
- **SMA:** SubMiniature Version A Connector

- **SNR:** Signal-to-Noise Ratio
- **SPI:** Serial Peripheral Interface
- **TGC:** Time Gain Compensation
- **T/R:** Transmit and Receive
- **UART:** Universal Asynchronous Receiver-Transmitter
- **VGA:** Variable Gain Amplifier

Tables

TABLE I

COMPARATIVE OVERVIEW OF PHASED ARRAY SYSTEMS AND SINGLE-ELEMENT PLATFORM

Engineering Parameter	Electronic Phased Array	Our Mechanical Scanning Platform
Acoustic Elements	64 to 256 piezoelectric elements	Single piezoelectric crystal (5 MHz DOBO 5P Φ10)
Transmitter Channels	64 to 256 synchronized HV pulsers	Single high-voltage pulser (MD1213 + TC6320)
Receiver Channels	64 to 256 parallel LNA/VGA stages	Single analog front end channel (AD8331)
Beamforming Method	Electronic delays and phase shifting	Physical Cartesian raster motion (stepper motors)
Hardware Complexity	Very high (specialized ASICs, high routing density)	Low to moderate (custom 2-layer PCB + Arty S7 FPGA)
System Cost	Thousands to tens of thousands of USD	Under 150 USD total electronic and structural BOM
Frame Rate	High (20 to 60 fps real-time)	Low (limited by mechanical scanning velocity)
Primary Suitability	In-vivo dynamic clinical imaging	Static phantoms, education and non-destructive testing

TABLE II

POWER RAIL ALLOCATIONS, REGULATING DEVICES AND TARGETED CIRCUIT LOADS

Voltage Rail	Regulator Component	Input Source	Assigned Subsystems
+100 V DC	U11 Boost Module + K1 Relay	+12 V Input	TC6320 high-voltage MOSFET pulser stage

+10.0 V DC	LM2940T-10.0 LDO (U7)	+12 V Input	MD1213 high-speed MOSFET gate driver
+6.5 V DC	XL7015 Buck Module (U10)	+12 V Input	Pre-regulation intermediate rail for 5 V LDO
+5.0 V DC	L7805CV Linear Regulator (U9)	+6.5 V Intermediate	AD8331 LNA/VGA analog supply and relay coil
+3.3 V DC	LD1117V33 LDO (U8)	+5.0 V Rail	AD9226 digital IO, MCP4821 DAC and logic interfaces

TABLE III

QUANTITATIVE PERFORMANCE PARAMETERS OF THE MECHANICAL ULTRASOUND SCANNER PLATFORM

System Specification	Measured / Characterized Value	Measurement Condition / Evaluation Method
Transducer Frequency	5.0 MHz center frequency	DOBO 5P Φ10 single-crystal immersion probe
Excitation Pulse Voltage	Approximately 95 V to 100 V	Measured at SMA connector CON1 into 50 Ohm load
Pulse Duration	100 ns to 120 ns	Half-period tuning for 5 MHz acoustic resonance
Receiver Dynamic Range	48 dB variable gain (14 to 62 dB total)	AD8331 LNA/VGA controlled by MCP4821 SPI DAC
Sampling Rate	20 MSPS (up to 65 MSPS)	AD9226 12-bit parallel ADC clocked by Arty S7 FPGA
Acoustic Axial Resolution	Approx. 0.35 mm (in water at 1480 m/s)	Evaluated from half-amplitude duration of envelope peak
Gantry Travel Envelope	200 mm x 150 mm	Active scan area within central plastic water container
Mechanical Repeatability	< 0.05 mm across successive sweeps	Dual lead screws on Y-axis with GT2 timing belt on X-axis
Host Interface	UART serial stream @ 921,600 baud	Digilent Arty S7 USB-UART bridge to Java desktop GUI

TABLE IV

BILL OF MATERIALS (BOM) OF KEY INTEGRATED CIRCUITS AND FUNCTIONAL COMPONENTS

Designator	Part Number / Model	Manufacturer	Package	Functional Description
U1	MD1213K6-G	Microchip	12-Lead QFN	High-speed dual MOSFET gate driver (10 V)
U2	TC6320TG-G	Microchip	8-Lead DFN	Complementary N- and P-channel HV MOSFET pair (200 V)
U3	MD0100	Microchip	3-Lead SOT-89	High-voltage bidirectional protection T/R switch
U4	AD8331ARQZ	Analog Devices	20-Lead QSOP	Ultralow noise preamplifier and VGA (48 dB range)
U5	MCP4821-E/SN	Microchip	8-Lead SOIC	12-bit SPI voltage DAC with internal reference (TGC)
U6	AD9226ASTZ	Analog Devices	48-Lead LQFP	12-bit parallel 65 MSPS analog-to-digital converter
U7	LM2940T-10.0	Texas Instruments	TO-220	Low-dropout 10.0 V linear regulator (gate driver supply)
U8	LD1117V33	STMicroelectronics	TO-220	Low-dropout 3.3 V linear regulator (logic and DAC supply)
U9	L7805CV	STMicroelectronics	TO-220	5.0 V positive linear voltage regulator (AD8331 supply)
U10	XL7015 Module	XLSEMI	Module	High-voltage step-down buck converter (12 V to 6.5 V)
U11	Boost Converter Module	Generic	Module	DC-DC step-up boost converter (12 V to 100 V DC)
K1	SRD-05VDC-SL-C	Songle	Through-hole	5 V DC electromechanical relay for HV safety isolation
D1, D2	BAV99	Nexperia	SOT-23	Dual high-speed switching diode clamps (transient protection)
D3, D4, D5	SS16	Vishay	SMA	1 A 60 V Schottky barrier diodes (reverse protection and flyback)
D6	B1100-13F	Diodes Inc.	SMA	1 A 100 V Schottky diode for high-voltage output protection
Q1	2N7002 / N-MOS	Various	SOT-23	N-channel logic-level MOSFET for relay coil switching

CON1	SMA Connector	Amphenol	Edge Mount	50 Ohm coaxial RF connector for ultrasonic transducer
CON2, 3, 4	2x6 PMOD Header	Samtec	Through-hole	Direct digital interface headers to Digilent Arty S7 FPGA
CON5	6-Pin Connector	Molex / JST	Through-hole	Stepper motor power (+12 V) and control signal header
Transducer	DOBO 5P Φ10	DOBO	Coaxial probe	5.0 MHz single-element immersion ultrasonic transducer

Figures

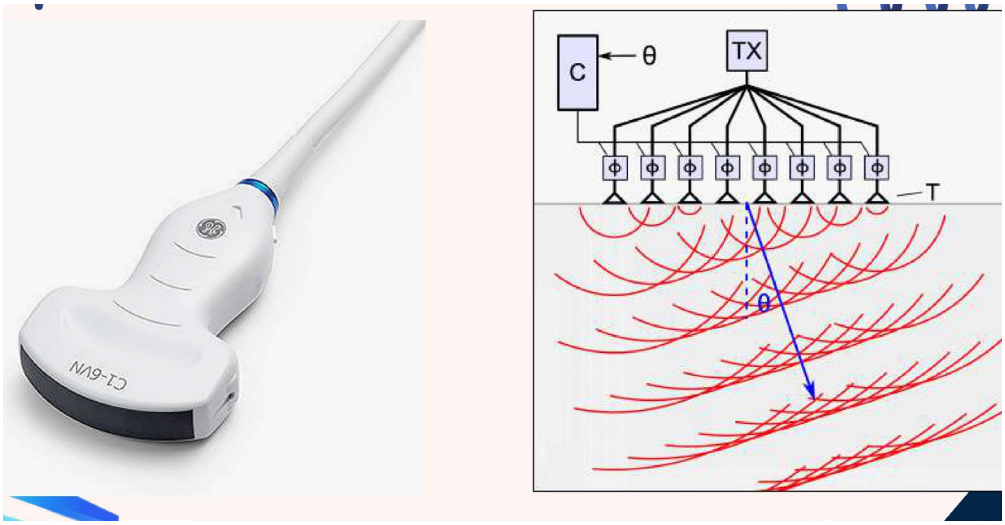


Fig. 1. *Commercial phased array transducer probe (left) and multichannel electronic beam steering network with delay lines (right).*



Fig. 2. *Single-element ultrasonic immersion transducer (DOBO 5P Φ10, 5 MHz center frequency, 10 mm crystal diameter).*

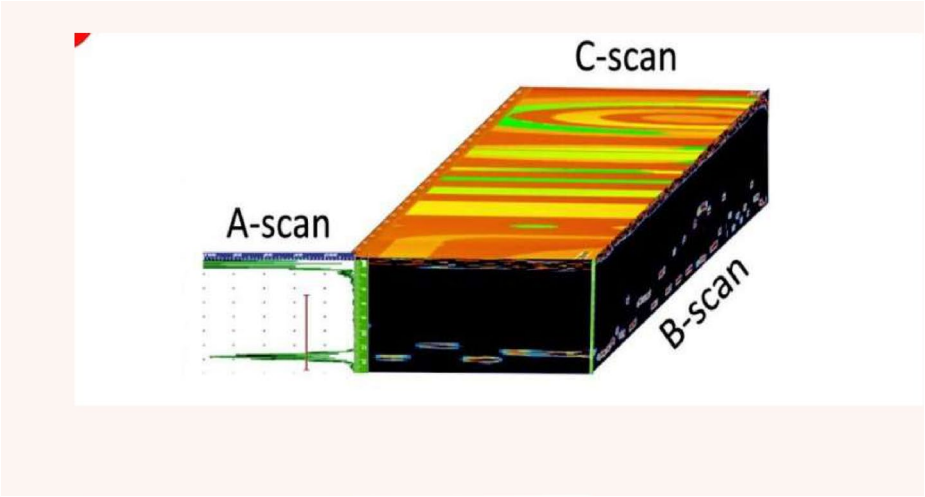


Fig. 3. *Three-dimensional geometric relationship between A-scan (1D echo trace), B-scan (2D cross-sectional slice) and C-scan (volumetric plane).*

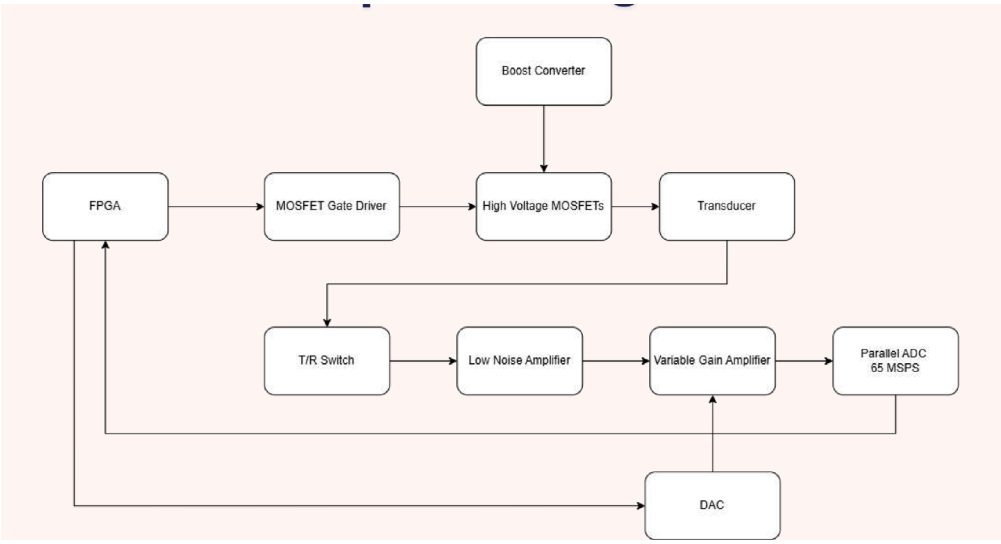


Fig. 4. *Simplified system architecture of the single-element ultrasound scanner, showing transmit path, receive path, FPGA and DAC control.*

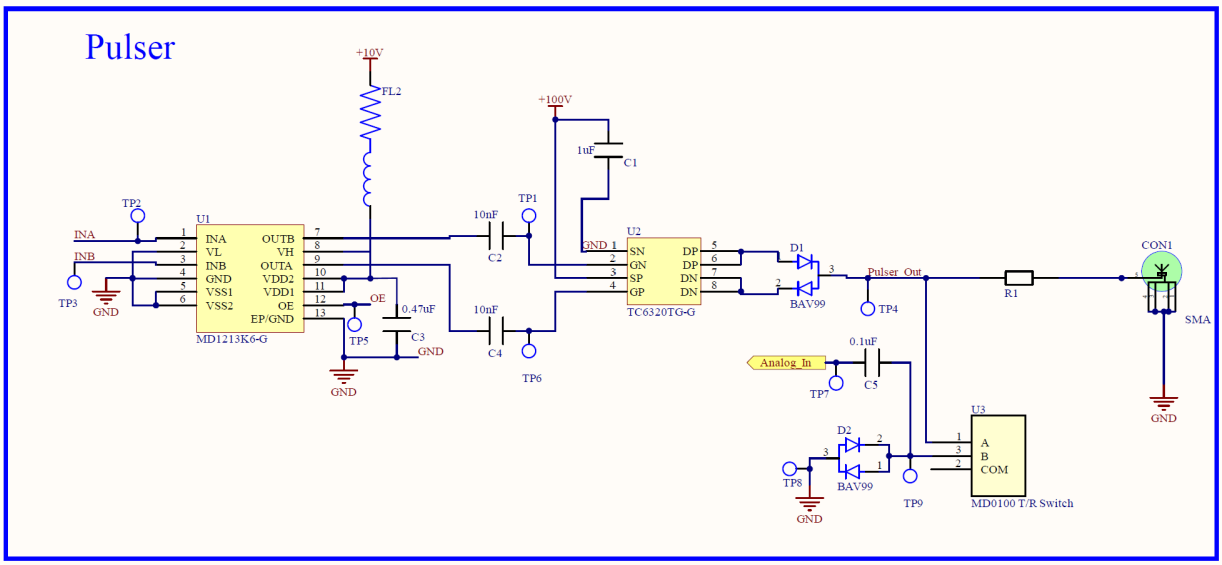


Fig. 5. *Altium schematic of the high-voltage pulser circuit, showing the MD1213 gate driver, TC6320 MOSFET pair and MD0100 T/R switch.*

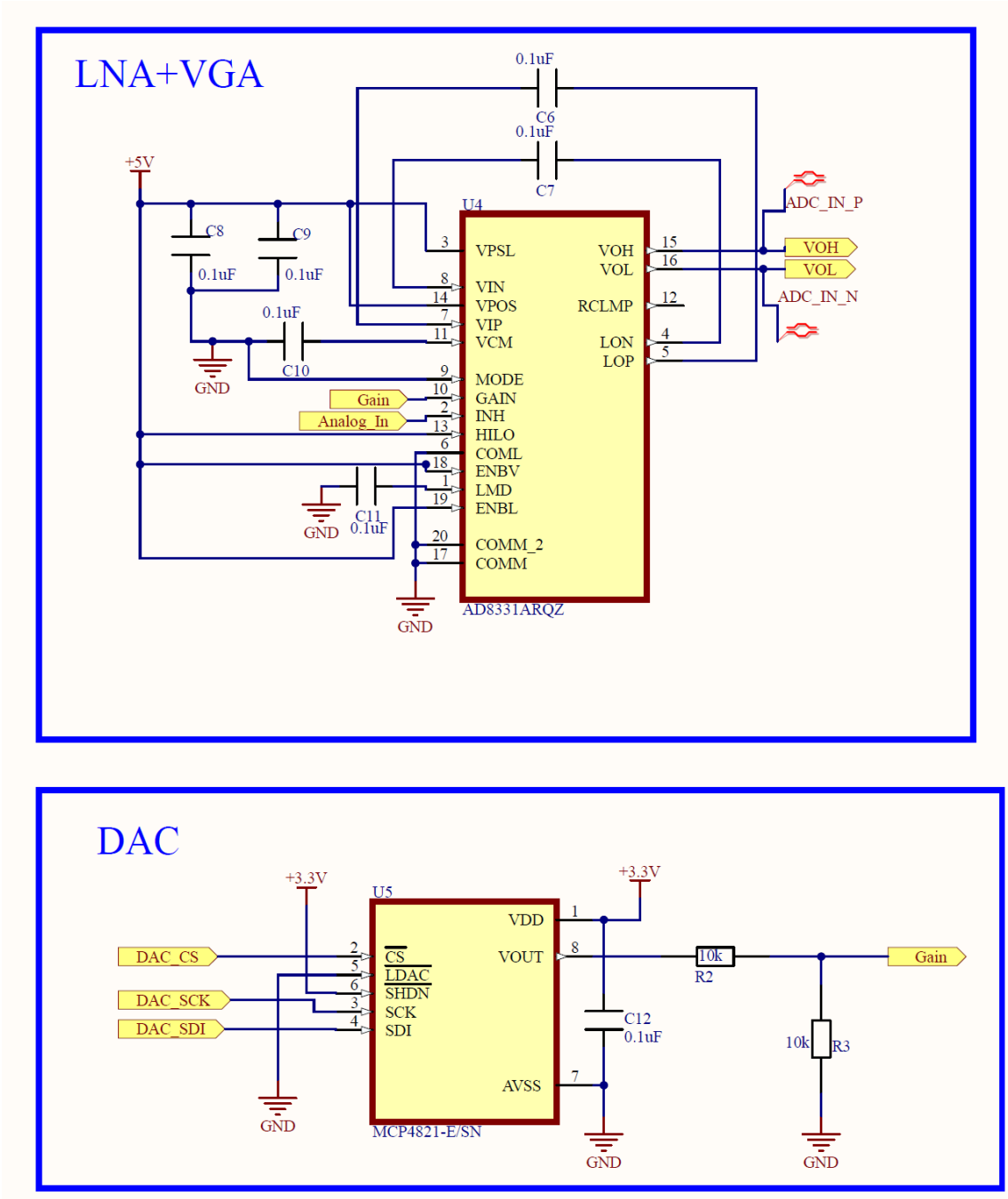


Fig. 6. *Altium schematic of the analog front end, showing the AD8331 LNA/VGA receiver and MCP4821 SPI DAC.*

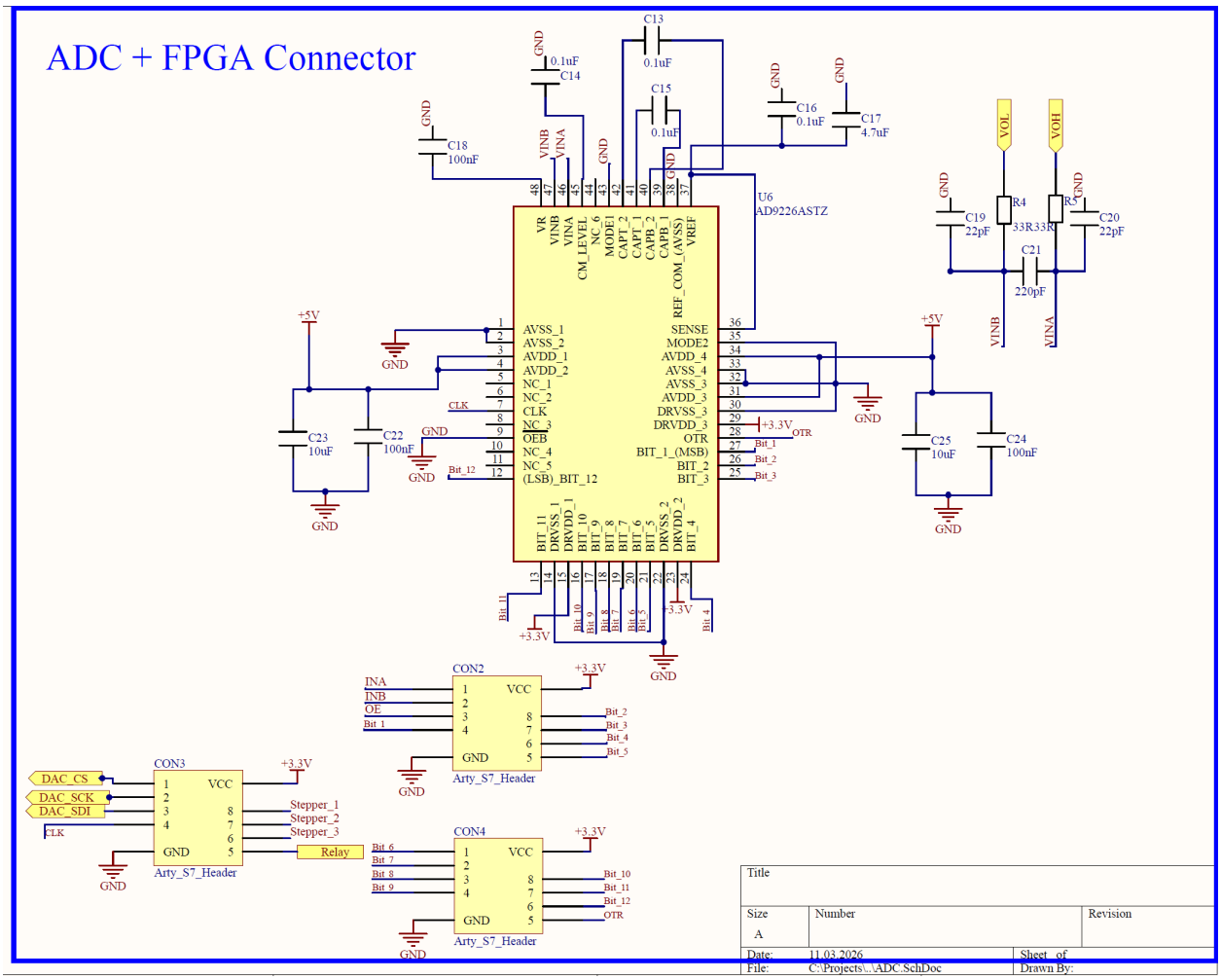


Fig. 7. *Altium schematic of the AD9226 12-bit 65 MSPS ADC and Arty S7 FPGA header connectors.*

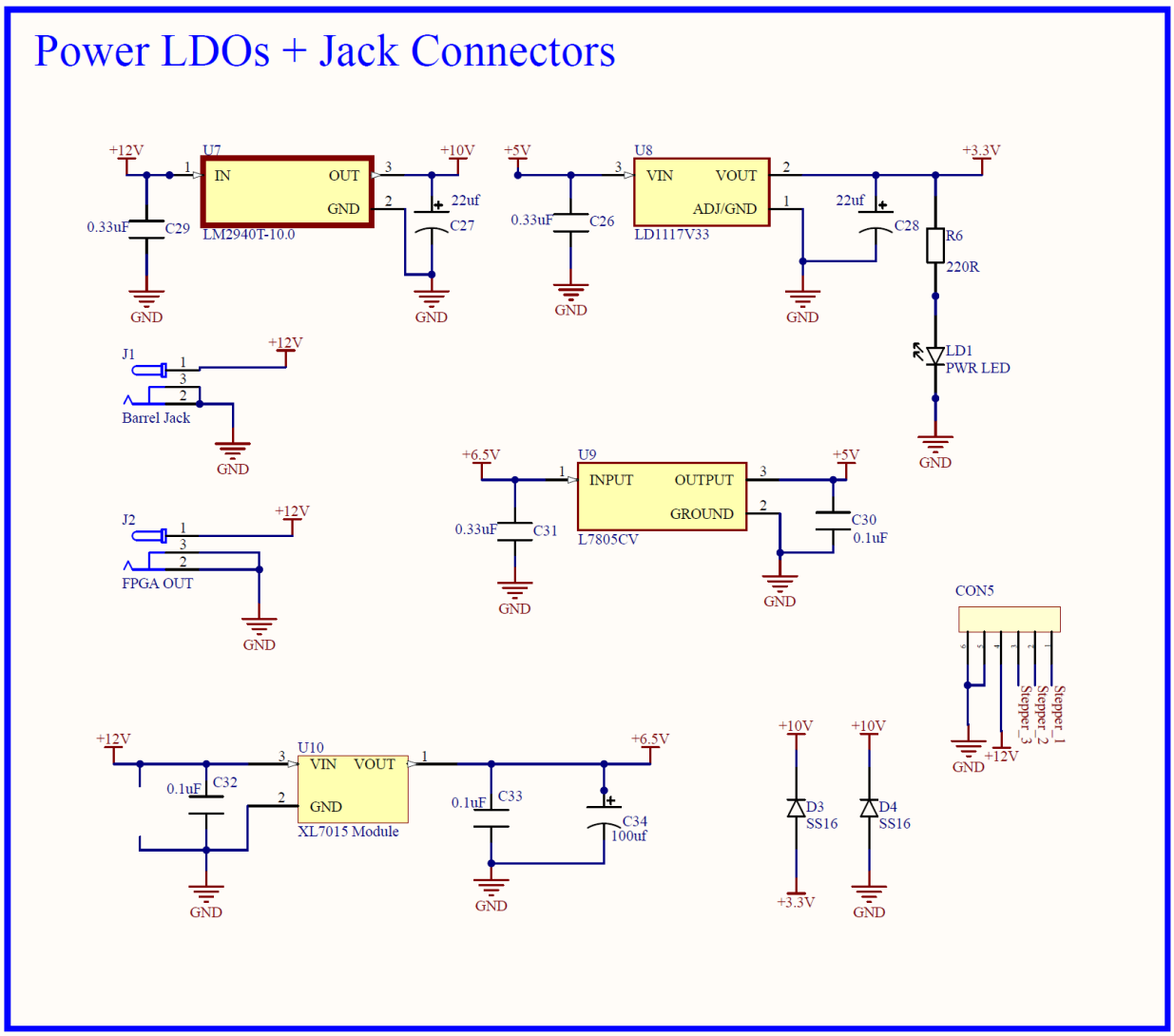


Fig. 8. *Altium schematic of the low-voltage power regulation network and stepper motor interface.*

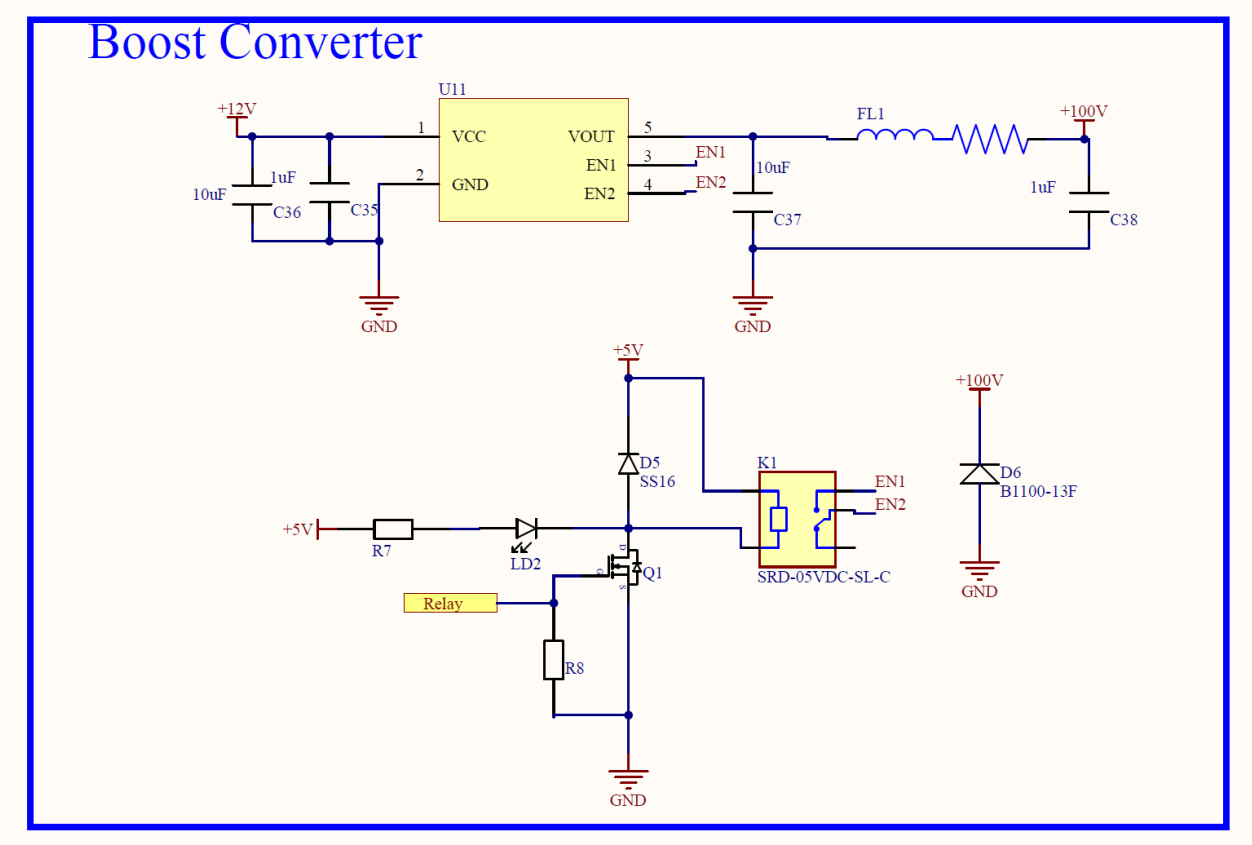
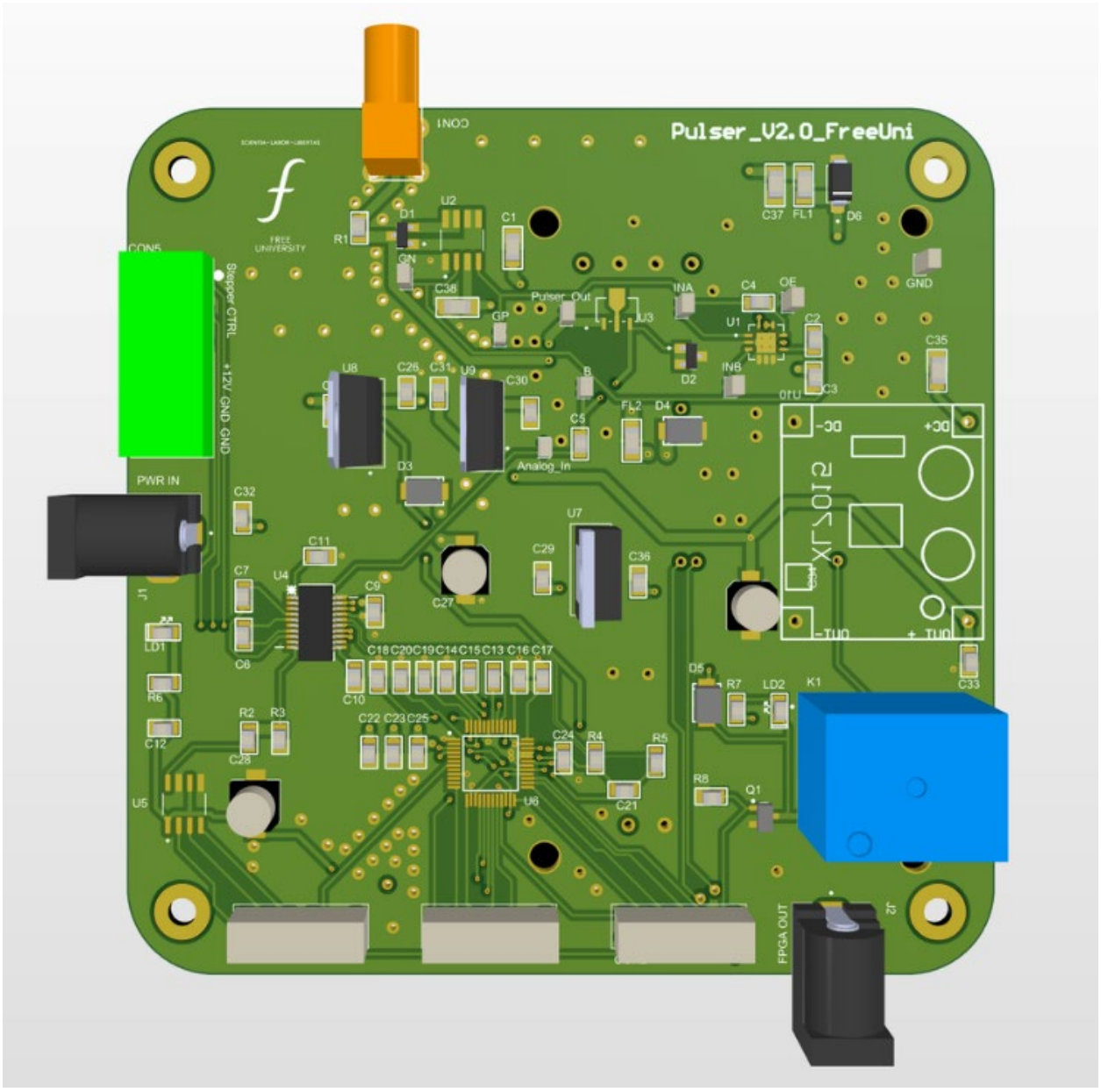


Fig. 9. *Altium schematic of the high-voltage boost converter circuit with relay isolation and MOSFET switch.*


Pulser_V2.0_FreeUni
INOO
FREE UNIVERSITY
CON5
Stepper CTRL
+12V GND GND
PWR IN
J1
J2
FPGA OUT
Pulser_Out
Analog_In
INA
INB
GP
GN
GND
U1
U2
U3
U4
U5
U6
U7
U8
U9
D1
D2
D3
D4
D5
D6
FL1
FL2
LD1
LD2
K1
Q1
R1
R2
R3
R4
R5
R6
R7
R8
C1
C2
C3
C4
C5
C6
C7
C8
C9
C10
C11
C12
C18 C20 C19 C14 C15 C13 C16 C17
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C22 C23 C25
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OE
-DC
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+TUO

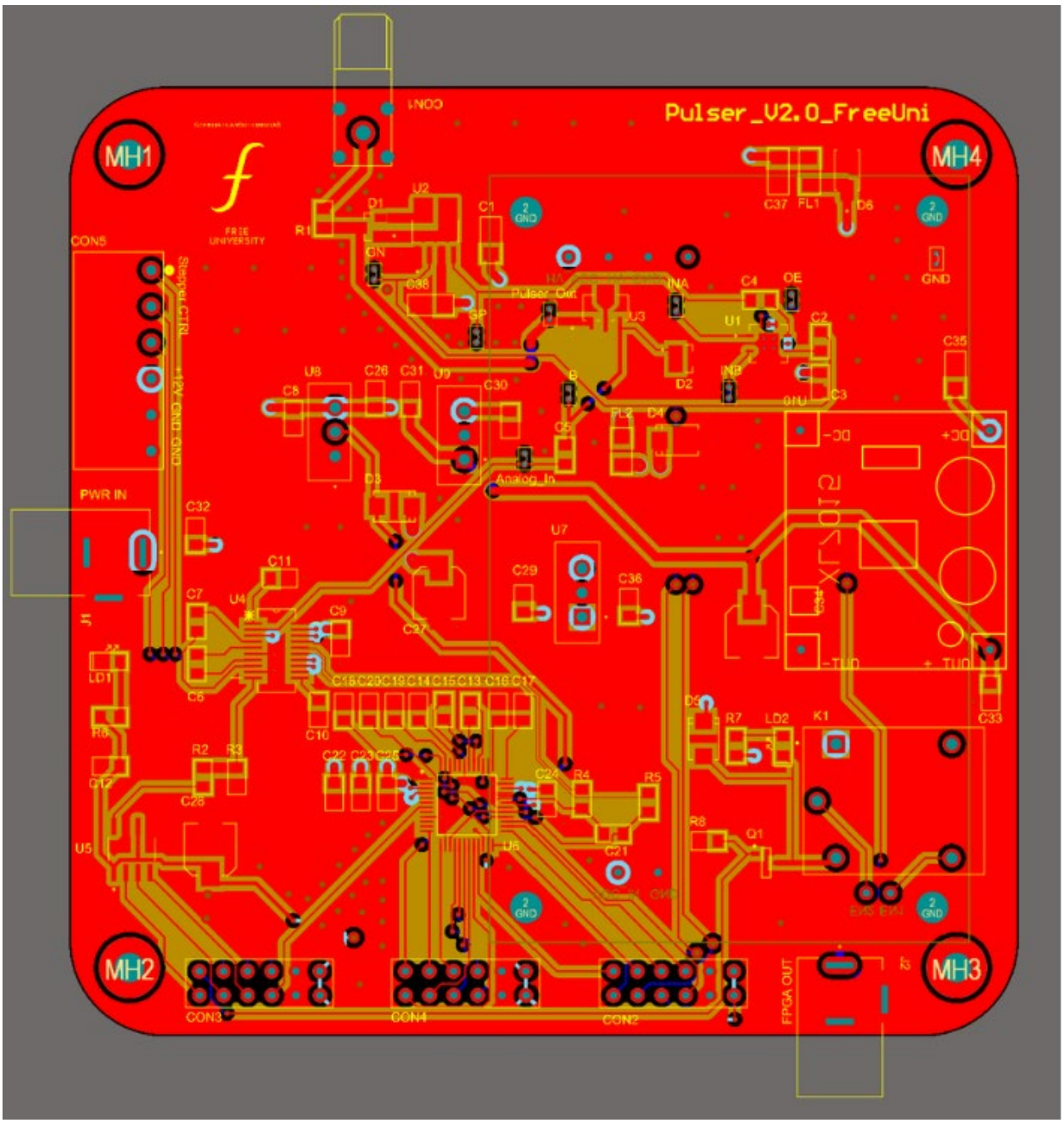


Fig. 10. *PCB layout, showing component placement* ***and*** *routing.*

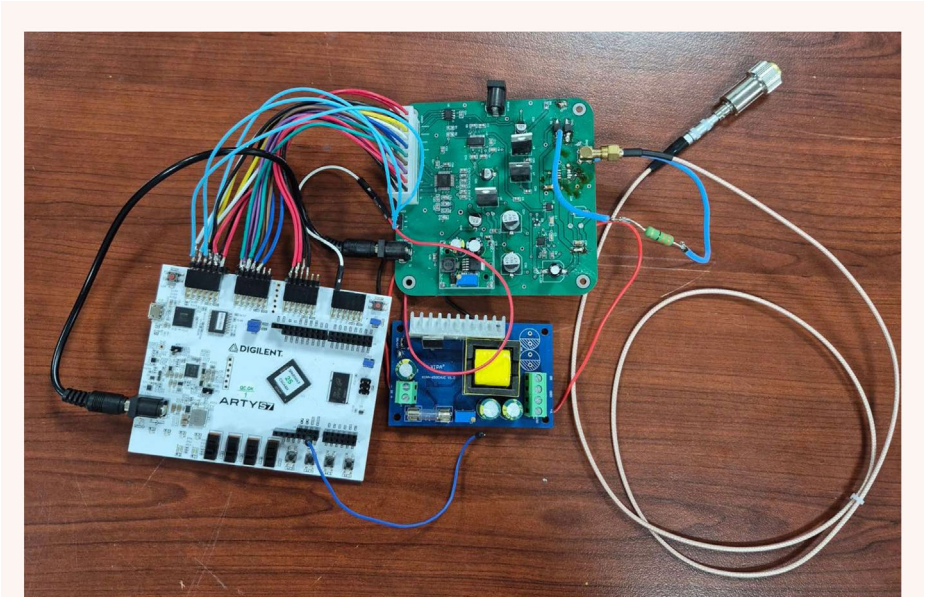


Fig. 11. *Initial hardware testbench setup (Version 1.0) with the Digilent Arty S7 FPGA board, prototype receiver board and boost module.*

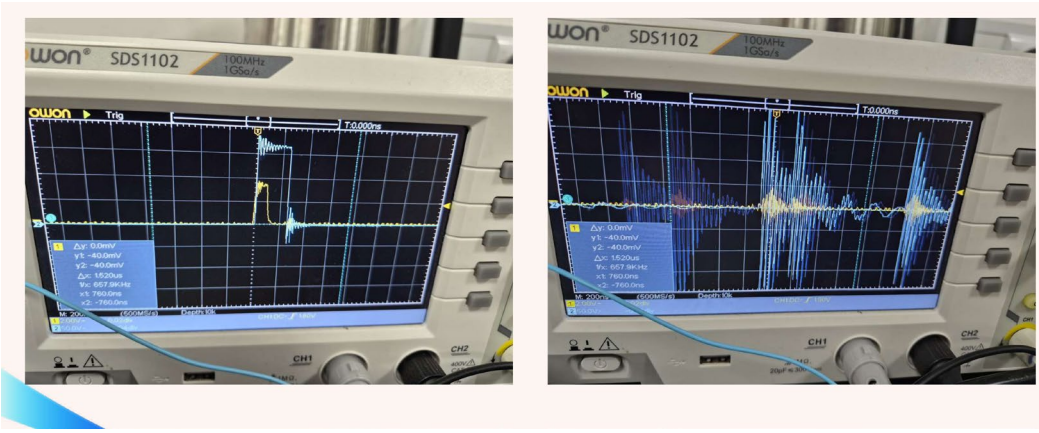


Fig. 12. *Oscilloscope captures of v1.0 issues: severe ringing on the high-voltage excitation pulse (left) and baseline noise on echo reception (right).*

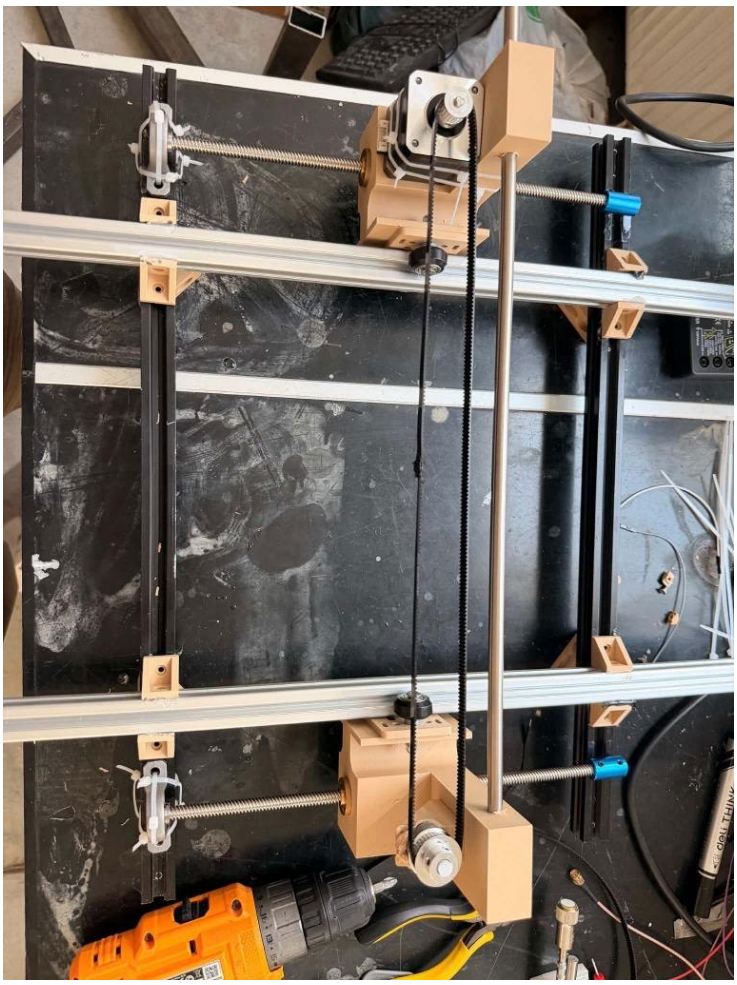


Fig. 13. *Motorized Cartesian gantry assembly showing aluminum extrusion rails, timing belt, dual lead screws and central container scanning bay.*

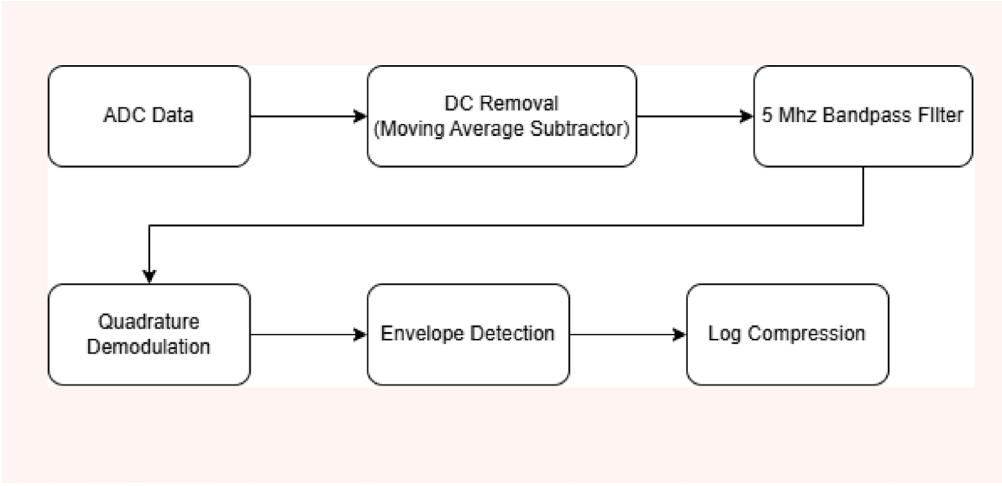


Fig. 14. *Digital Signal Processing (DSP) pipeline flow diagram from raw ADC samples to log-compressed display values.*

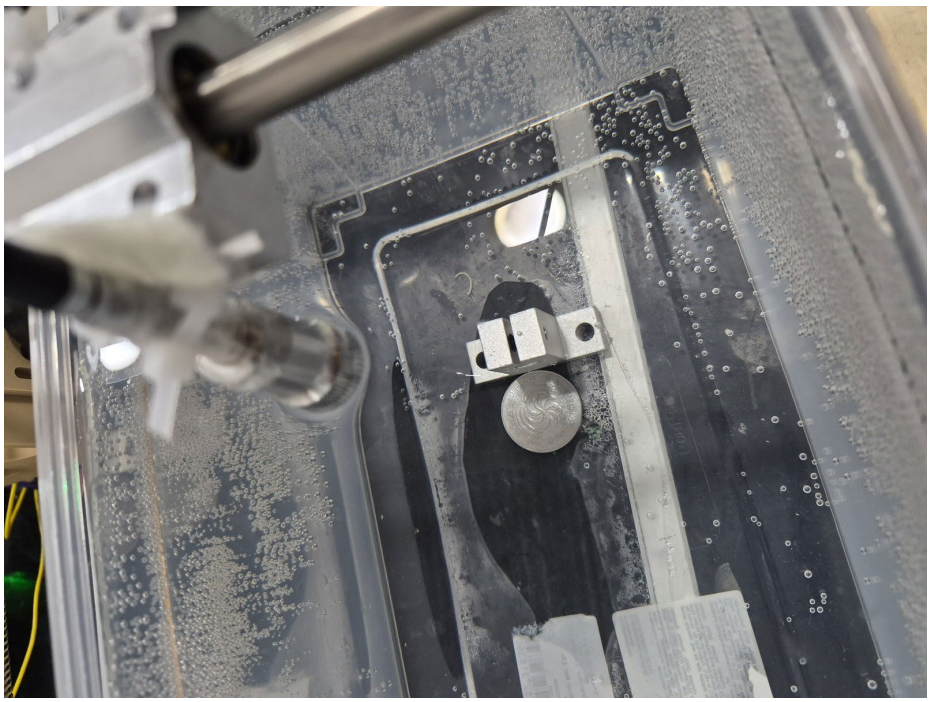


Fig. 15. *Experimental test setup for metal specimen scanning, showing the 5 MHz immersion transducer mounted on the motorized Cartesian gantry above the submerged coin target in the coupling water tank.*

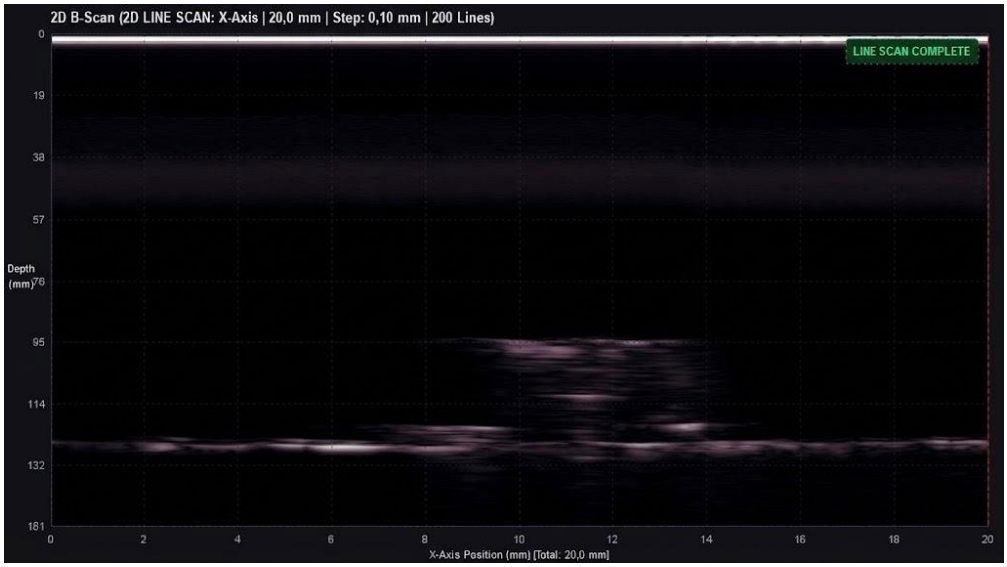


Fig. 16. *Two-dimensional cross-sectional B-scan ultrasound echogram of the submerged metal coin target, showing high-amplitude echo reflection bands at the solid-liquid boundary.*

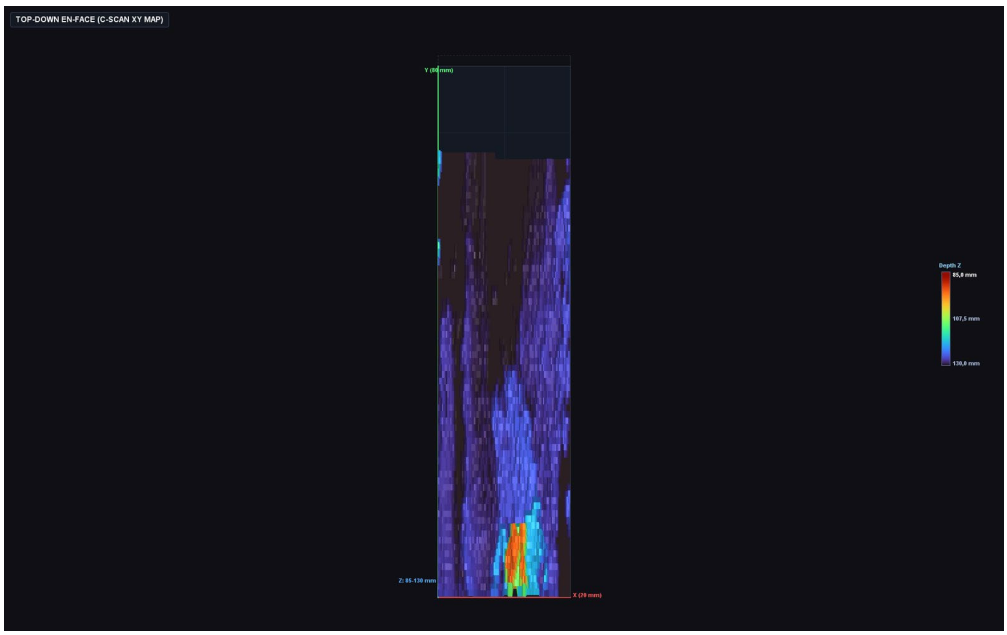


Fig. 17. *Reconstructed en-face C-scan top-down acoustic projection of the metal coin specimen, resolving the circular outer contour and surface relief features across the scanning grid.*

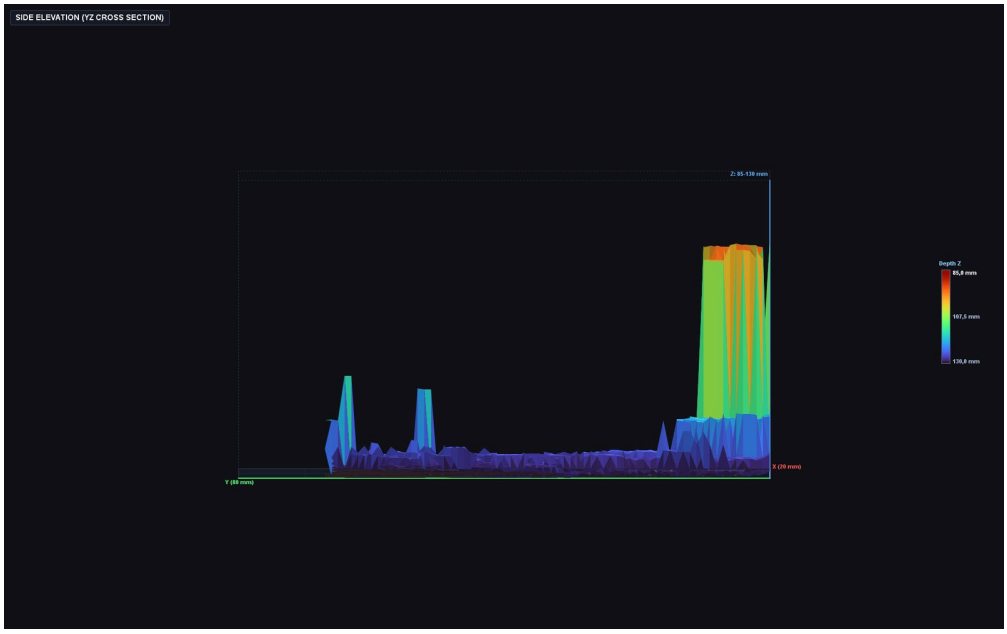


Fig. 18. *Oblique three-dimensional surface topography rendering of the scanned metal coin target generated by the Java host visualization engine, displaying elevation-coded colormapping and directional hill-shading.*



Fig. 19. *Experimental bench setup for submerged machine screw scanning, showing the specimen positioned within the central water container beneath the motorized raster gantry.*

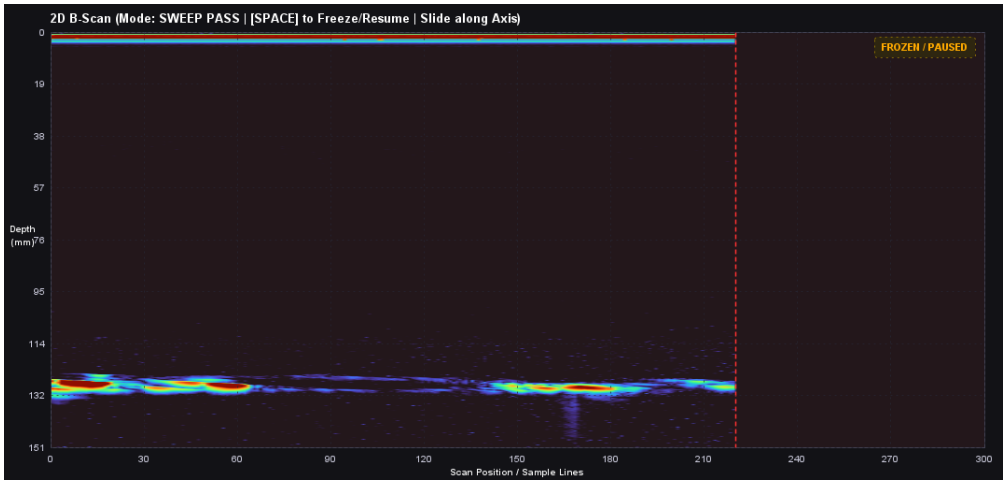


Fig. 20. *Two-dimensional B-scan ultrasound cross-sectional slice across the submerged machine screw, capturing distinct echo arrival times from the screw head and threaded shaft.*

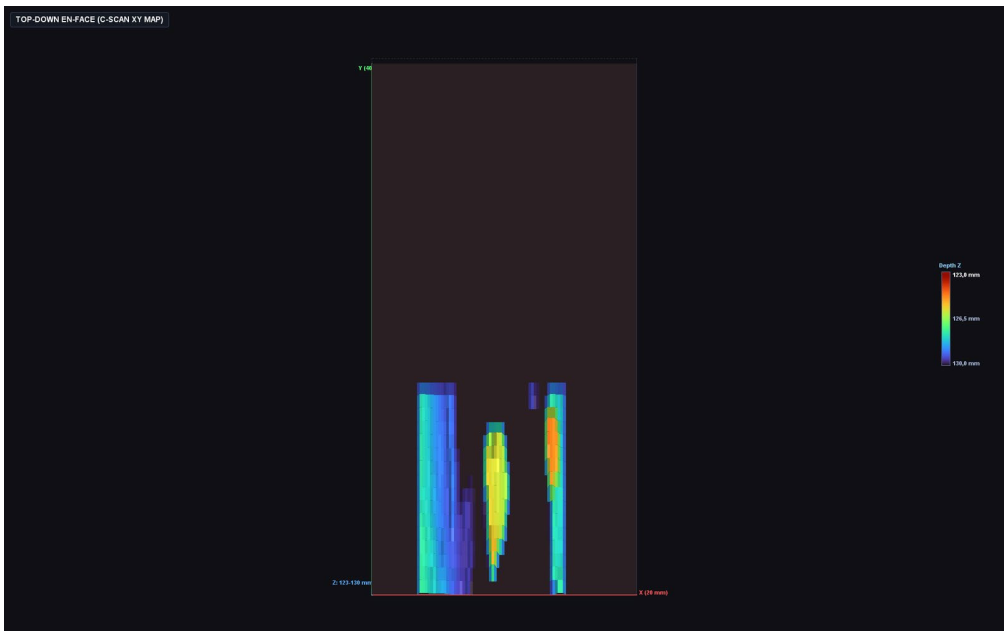


Fig. 21. *Reconstructed en-face C-scan top-down acoustic intensity map of the submerged machine screw, clearly resolving the circular drive head, elongated cylindrical body and threaded pitch profile.*

1. Introduction

1.1 Project Background and Clinical Context

Ultrasonic imaging is an indispensable tool in modern healthcare and non-destructive material testing [1], [2]. Unlike ionizing imaging modalities such as X-ray radiography or computed tomography, acoustic imaging uses high-frequency mechanical sound waves that travel through media and biological tissues safely without harmful radiation. When an acoustic wave propagates across boundaries separating media with differing acoustic impedances, a fraction of the wave energy reflects back toward the source as an echo. By measuring the arrival time and amplitude of returning echoes, an imaging system reconstructs internal maps of structural boundaries, density transitions and material discontinuities.

In clinical practice, cross-sectional brightness mode (B-scan) ultrasound provides real-time diagnostic evaluation of abdominal organs, cardiovascular structures and musculoskeletal tissues. Acoustic reflections from anatomical interfaces are rendered as varying grayscale intensity levels, where strongly reflective surfaces appear bright and fluid-filled cavities appear dark.

1.2 Phased Arrays vs Single-Element Scanning

Conventional diagnostic ultrasound platforms rely heavily on multi-element phased array transducers, such as the commercial curved convex array probe illustrated in Fig. 1. In a phased array, dozens or hundreds of piezoelectric crystal elements are arranged side by side in a compact housing. By exciting these elements with individual delay lines and applying calculated phase shifts, the system steers and focuses the ultrasonic beam electronically across an angular sector without moving the probe physically.

While electronic beam steering provides rapid frame rates suitable for moving organs, it introduces substantial electronic complexity. Each individual piezoelectric element requires a dedicated high-voltage transmitter channel, an ultra-low-noise preamplifier, a programmable analog delay stage and high-speed analog-to-digital conversion. A typical 64-channel or 128-channel clinical machine requires specialized multi-channel application-specific integrated circuits (ASICs), dense multi-layer printed circuit boards and expensive custom transducer

arrays. For university laboratories, student coursework and resource-limited engineering initiatives, acquiring or custom-building multi-channel phased array hardware is economically impractical.

To address this accessibility barrier, our project investigated an automated single-element mechanical scanning approach. Instead of steering an acoustic beam electronically using hundreds of channels, we use a single high-performance piezoelectric immersion transducer translated mechanically across the target area by a motorized Cartesian gantry. The transducer chosen for our implementation is the DOBO 5P Φ10 immersion probe shown in Fig. 2. This probe features a single piezoelectric crystal element operating at a center frequency of 5 MHz with an active crystal diameter of 10 mm, terminated in a standard coaxial RF connector.

Table I highlights the primary engineering differences between conventional multi-element phased arrays and our single-element mechanical scanning platform.

1.3 Ultrasound Imaging Modes: A-Scan, B-Scan and C-Scan

The dimensional hierarchy of ultrasound data acquisition forms the theoretical basis of our scanning system [1], [3]. Acoustic measurements are classified into three geometric representations, depicted in Fig. 3, utilizing the single-element DOBO 5P Φ10 immersion transducer [4]:

- **A-Scan (Amplitude Mode):** A one-dimensional radio frequency (RF) time-domain waveform recorded from a single pulse transmission. The horizontal axis represents round-trip acoustic travel time (proportional to target depth given the speed of sound in the propagation medium, approximately 1480 m/s in water), while the vertical axis represents the amplitude of reflected acoustic pressure waves.
- **B-Scan (Brightness Mode):** A two-dimensional cross-sectional slice. A sequence of adjacent A-scan lines are acquired while translating the transducer along a linear axis. The envelope of each A-scan is computed and mapped to grayscale pixel brightness, producing a two-dimensional image where acoustic impedance boundaries appear as bright reflective contours against a dark background.

- **C-Scan (Constant Depth or Volumetric Mode):** A two-dimensional slice parallel to the scanning plane at a specified depth or a complete three-dimensional voxel volume formed by acquiring a raster grid of B-scans across two spatial axes (X and Y).

1.4 Objectives and Project Scope

The primary goal of this senior project was to design, construct and validate an automated, low-cost ultrasound scanning apparatus capable of generating cross-sectional B-scans and volumetric representations of submerged targets using a single transducer. The scope of work encompassed the following engineering tasks:

- **High Voltage Pulser Design:** Design an electronic pulser capable of delivering unipolar high-voltage pulses of approximately 100 V to drive the 5 MHz piezoelectric crystal into resonance.
- **T/R Switch and Protection:** Implement an active high-voltage transmit/receive switch and fast clamping diodes to isolate sensitive receiver amplifiers from 100 V transmit bursts while passing sub-millivolt echo reflections.
- **Low Noise Analog Front End:** Construct a receiver using an ultra-low-noise preamplifier and variable gain amplifier with Time Gain Compensation (TGC) to offset acoustic attenuation over depth.
- **High Speed Digitization:** Sample returning RF echo signals using a 12-bit parallel analog-to-digital converter operating at 65 MSPS, interfaced directly to an FPGA development board.
- **FPGA Control and DSP:** Implement digital signal processing routines on a Digilent Arty S7 FPGA board, including DC offset removal, digital bandpass filtering and quadrature demodulation for envelope extraction.
- **Cartesian Motion Platform:** Fabricate a rigid Cartesian gantry using aluminum extrusions and 3D printed components to translate the transducer across a central water container holding the test specimen.
- **Host Visualization Software:** Develop a desktop application in Java to manage serial communication, map acoustic data to spatial coordinates and render B-scan images in real time.

2. Problem Formulation and Proposed Methodology

2.1 System Architecture Overview

The complete architecture of our ultrasound scanner platform is illustrated in the simplified system block diagram in Fig. 4. The architecture comprises a high-voltage transmit path, a protected low-noise receive conditioning chain, digital acquisition logic and mechanical motion synchronization.

Operation begins with the Digilent Arty S7 FPGA development board [12], built around an AMD Xilinx Spartan-7 FPGA [13], which acts as the master system controller. When an acoustic acquisition line is initiated, the FPGA issues precise digital timing pulses (INA, INB and OE) to a high-voltage MOSFET gate driver. The gate driver switches a pair of high-voltage complementary MOSFETs connected to a +100 V DC bus supplied by an isolated boost converter. This generates a sharp high-voltage negative-going pulse that excites the single-element immersion transducer through a coaxial SMA connection.

The resulting acoustic pressure wave travels into the immersion fluid (water) and reflects off target interfaces. Returning acoustic echoes strike the transducer, generating faint electrical signals on the order of microvolts to millivolts. A high-voltage Transmit/Receive (T/R) switch protects the sensitive analog electronics by blocking the 100 V transmit pulse, while conducting weak returning echoes into a Low Noise Amplifier (LNA) and Variable Gain Amplifier (VGA). To counteract signal attenuation caused by acoustic absorption over depth, a Digital-to-Analog Converter (DAC) driven by the FPGA generates a rising voltage ramp that sweeps the VGA gain during the listening window.

The amplified differential echo signal is digitized by a 12-bit parallel ADC operating at up to 65 MSPS. The digitized data stream is ingested by the FPGA, processed through digital filters, demodulated and transmitted via UART to the host computer. Simultaneously, the FPGA outputs step and direction pulses to motor drivers controlling the Cartesian gantry, synchronizing mechanical displacement with acoustic line acquisition.

2.2 Transceiver Electronics and Circuit Design

The electronics subsystem was designed in Altium Designer across five schematic sheets and integrated onto a custom two-layer PCB titled Pulser_V2.0_FreeUni. All schematics and the PCB layout are presented in the Figures section (Fig. 5 through Fig. 10).

2.2.1 High Voltage Pulser and T/R Switch

The pulser schematic, shown in Fig. 5, generates the high-voltage excitation pulse that drives the transducer into acoustic vibration, while protecting the receive channel from damaging voltages.

The circuit is built around U1, an MD1213K6-G high-speed dual MOSFET driver [5]. The driver accepts low-voltage logic inputs (INA, INB and OE) from the FPGA at 3.3 V CMOS levels on pin 2 (VL). The driver is powered from a dedicated +10 V supply rail (pins 8, 10 and 11) to achieve fast gate charging and discharging. The driver outputs OUTA (pin 9) and OUTB (pin 7) are AC-coupled through 10 nF capacitors C2 and C4 to the gates of U2, a TC6320TG-G integrated complementary N-channel and P-channel high-voltage MOSFET pair [6].

The drain terminals of the TC6320 pair are tied together at net Pulser_Out, connected to the +100 V boost converter rail through bypass capacitor C1 (1 uF) and inductor filter FL2. By alternating the conduction of the N-channel and P-channel transistors for approximately 100 ns, net Pulser_Out delivers a short high-energy impulse to the transducer connected at SMA connector CON1.

To protect the analog receiver from this +100 V excitation, net Pulser_Out is isolated from net Analog_In by U3, an MD0100 high-voltage protection T/R switch [7]. The MD0100 is a normally-closed, bidirectional depletion-mode switch that turns off when the voltage across its terminals exceeds approximately ±2 V, presenting a high impedance of over 200 kOhm that blocks high-voltage spikes. In addition, dual back-to-back Schottky diode pairs D1 and D2 (Nexperia BAV99 [8]) clamp any residual turn-on transient spikes to within safe logic rails, ensuring that node Analog_In never experiences voltages outside the -0.7 V to +0.7 V range.

2.2.2 Analog Front End (AFE) and Time Gain Compensation (TGC)

The returning echo signals entering net Analog_In have amplitudes ranging from a few microvolts up to several tens of millivolts. The analog front end, shown in Fig. 6, provides amplification, variable gain control and differential conversion.

The analog receiver is built around U4, an Analog Devices AD8331ARQZ ultra-low-noise preamplifier and variable gain amplifier [9]. The single-ended acoustic echo signal from net Analog_In passes through AC-coupling capacitor C10 (0.1 uF) into the low-noise preamplifier input pin 2 (INH). The AD8331 preamplifier achieves an input voltage noise spectral density of only 0.74 nV/sqrt(Hz), which is critical for preserving weak acoustic reflections from deep targets.

The output of the internal LNA is routed differentially to pins 4 (LON) and 5 (LOP), where external AC coupling through 0.1 uF capacitors C6 and C7 feeds the integrated Variable Gain Amplifier (VGA) inputs VIP (pin 7) and VIN (pin 8). The VGA provides a gain adjustment range of 48 dB, controlled by the voltage applied to pin 10 (GAIN). Pin 9 (MODE) is grounded to select a positive gain slope of 50 dB/V, while pin 13 (HILO) selects the high-gain operating range (14 dB to 62 dB total system gain).

To implement Time Gain Compensation (TGC), U5, a Microchip MCP4821-E/SN 12-bit SPI voltage DAC [10], generates an analog voltage ramp. The DAC is powered from +3.3 V and driven by the FPGA over three dedicated lines: DAC_CS, DAC_SCK and DAC_SDI. The DAC voltage output on pin 8 (VOUT) passes through a precision 10 kOhm voltage divider (R2 and R3) to provide a 0 V to 1.0 V control signal on net Gain. At the start of each acoustic transmit event, the FPGA commands the DAC to its minimum voltage. As the sound wave penetrates deeper into the medium, the FPGA ramps up the DAC code over the listening period (typically 50 to 100 microseconds), progressively increasing VGA amplification to compensate for acoustic attenuation.

The amplified differential outputs of the AD8331 appear on pins 15 (VOH) and 16 (VOL), providing a balanced differential signal referenced to internal common-mode level VCM (pin 11).

2.2.3 Digitization and FPGA Interface

The differential outputs VOH and VOL are converted to digital words at high speed to reconstruct the 5 MHz acoustic carrier. The digitization circuit, shown in Fig. 7, uses an Analog Devices AD9226ASTZ 12-bit parallel ADC (U6) [11].

The AD9226 is configured for true differential input mode. Differential nets VOH and VOL pass through a symmetrical low-pass RC reconstruction filter composed of two 33 Ohm series damping resistors (R4 and R5), a 220 pF differential capacitor (C21) and two 22 pF capacitors to ground (C19 and C20). This filter suppresses high-frequency broadband noise above 15 MHz before the signal enters analog input pins 46 (VINA) and 47 (VINB).

The ADC operates at a conversion clock frequency of 50 MHz or 65 MHz provided by the FPGA on net CLK (pin 7). The digitized samples appear on a 12-bit parallel bus, spanning from pin 27 (Bit_1, MSB) down to pin 12 (Bit_12, LSB). Pin 28 (OTR) provides an out-of-range flag that alerts the FPGA if an acoustic reflection clips the analog input range.

To interface directly with the Digilent Arty S7 Spartan-7 FPGA board, the PCB incorporates three 2x6 header connectors (CON2, CON3 and CON4) matching standard PMOD pinouts:

- **CON2:** Connects transmit control signals INA, INB and OE, along with upper ADC data bits Bit_1 through Bit_5.
- **CON3:** Carries the SPI DAC control lines (DAC_CS, DAC_SCK and DAC_SDI), ADC clock (CLK), relay control (Relay) and three stepper motor control signals (Stepper_1, Stepper_2 and Stepper_3).
- **CON4:** Transfers lower ADC data bits Bit_6 through Bit_12 and the out-of-range flag (OTR).

2.2.4 Power Management and High Voltage Generation

The ultrasound scanner electronics require multiple distinct power supply rails with rigorous low-noise isolation, shown in the power distribution schematic in Fig. 8 and the boost converter schematic in Fig. 9.

The system is powered from an external +12 V DC wall adapter through barrel jack J1, with an auxiliary pass-through jack J2 for powering the FPGA board. Because the AD8331

preamplifier and AD9226 analog core are sensitive to switching regulator noise, linear regulators (LDOs) are used for all sensitive analog nodes:

- +6.5 V Intermediate Rail: An XL7015 buck regulator module (U10) [17] steps the incoming +12 V down to +6.5 V with high efficiency, reducing thermal dissipation in downstream regulators.
- +5.0 V Analog Rail: An L7805CV linear regulator (U9) [16] steps +6.5 V down to a quiet +5.0 V rail, powering the AD8331 LNA/VGA (U4) and the relay coil.
- +10.0 V Gate Driver Rail: An LM2940T-10.0 low-dropout regulator (U7) [14] steps +12 V down to clean +10.0 V, powering the MD1213 MOSFET driver (U1) to ensure fast gate switching.
- +3.3 V Digital Rail: An STMicroelectronics LD1117V33 linear regulator (U8) [15] steps +5.0 V down to +3.3 V, supplying the MCP4821 DAC, the AD9226 digital IO stage and FPGA pull-up networks.

High voltage for transducer excitation is produced by U11, an adjustable DC-DC boost converter module that steps +12 V up to +100 V DC. For safety during initialization and debugging, the high-voltage output is gated through K1, a Songle SRD-05VDC-SL-C electromechanical relay [18]. The relay coil is switched by N-channel MOSFET Q1 driven by the FPGA signal Relay through gate resistor R8 (10 kOhm). Diode D5 (SS16) suppresses inductive turn-off flyback and indicator LED LD2 provides a visual warning whenever the +100 V rail is active. Diode D6 (B1100-13F, 100 V 1 A Schottky [19]) blocks reverse current into the boost converter output. The complete power rail assignments are listed in Table II.

2.2.5 Custom Printed Circuit Board (PCB) Layout

The physical PCB layout, shown in Fig. 10, was completed in Altium Designer [23] on a two-layer FR4 board measuring 120 mm by 120 mm with rounded mounting corners and the Free University logo etched in copper.

The layout strictly segregates high-voltage noisy switching lines from sensitive analog receiver traces. The +100 V boost converter section and TC6320 pulser MOSFETs are located along the upper half of the board, directly adjacent to the SMA transducer connector (CON1) to

minimize trace length and parasitic loop inductance. The lower half of the board houses the AD8331 LNA/VGA, the MCP4821 DAC and the AD9226 ADC.

Grounding is handled through solid ground copper pours on both layers, stitched together with vias around high-current switching loops. Differential traces routing from AD8331 outputs VOH and VOL to AD9226 inputs VINA and VINB are routed as length-matched pairs shielded by adjacent ground pours to eliminate common-mode noise coupling.

2.3 Hardware Evolution and Lessons from Version 1.0

Before manufacturing the unified PCB, an initial proof-of-concept prototype (Version 1.0) was assembled on a laboratory test bench to validate pulse transmission, echo reception and FPGA acquisition. Fig. 11 shows this first-version experimental setup, consisting of a preliminary green receiver board, a separate blue boost converter module, the Digilent Arty S7 FPGA board and the DOBO 5 MHz transducer connected to a water test tank.

While prototype v1.0 proved that acoustic pulse-echo signals could be generated and digitized, laboratory bench measurements using an oscilloscope revealed significant signal integrity problems, illustrated in the captured oscilloscope traces in Fig. 12.

As seen in the left oscilloscope capture of Fig. 12, the high-voltage excitation pulse suffered from severe inductive ringing and an extended negative undershoot tail exceeding 500 ns. This ringing was traced to parasitic inductance in the flying jumper wires connecting the external boost converter and gate driver to the transducer. The long ringing obscured acoustic reflections originating from shallow targets, creating a large acoustic dead zone.

Furthermore, as seen in the right oscilloscope capture of Fig. 12, the receive channel exhibited excessive baseline noise and interference spikes during the listening window. Unshielded ribbon cables between the receiver board and the Arty S7 FPGA picked up radiated EMI from the boost converter switching circuit, while shared ground paths modulated the LNA input reference.

These findings led directly to the design revisions in Version 2.0: integrating all subsystems onto a single PCB, replacing flying jumper wires with direct PMOD headers, adding dedicated ground planes, placing localized decoupling capacitors directly at IC power pins and incorporating an RC damping filter across the ADC differential input.

2.4 Mechanical Cartesian Scanner

To generate two-dimensional B-scan slices and volumetric scans from a single transducer, we built an automated Cartesian gantry platform, shown in Fig. 13.

The scanner chassis is constructed from structural aluminum extrusion profiles: black anodized base rails provide rigid foundation support, while silver cross-beams form the linear guide tracks. All motion mounts, carriage blocks, motor brackets and bearing housings were custom-designed and 3D printed from high-strength PLA filament (visible as tan-colored structural parts in Fig. 13).

Mechanical motion is distributed across two primary axes:

- X-Axis (Fast Scan Axis): Driven by a single NEMA 17 bipolar stepper motor driven by an Allegro A4988 microstepping driver [20] and coupled to a continuous GT2 timing belt and idler pulley. The timing belt drive allows rapid linear sweeps across the scanning line without backlash.
- **Y-Axis (Slow Indexing Axis):** Driven by dual synchronized lead screws (threaded rods with anti-backlash brass nuts) supported on precision smooth guide rods. Using dual synchronized lead screws driven from opposite sides prevents mechanical racking or angular skewing of the carriage across its wide span.

As shown in Fig. 13, the central bay of the gantry accommodates a rectangular transparent plastic water container. The immersion transducer is mounted facing downward from the overhead carriage into the water container. Targets to be imaged (such as structural phantoms, layered materials and embedded geometries) are placed submerged at the bottom of the container. Because water provides excellent acoustic impedance matching to human soft tissue and biological phantoms, ultrasound energy couples efficiently into the submerged target without the acoustic shadowing typical of air gaps.

2.5 Digital Signal Processing (DSP) Pipeline

Once the returning acoustic echo is amplified and digitized by the AD9226 ADC at 50/65 MSPS, the raw digital RF data stream enters the digital signal processing pipeline. Fig. 14

outlines the complete DSP architecture, divided between the FPGA firmware, synthesized in AMD Xilinx Vivado [24] and the host desktop application.

The processing pipeline executes through the following stages:

- **1. ADC Data Ingestion:** The FPGA captures 12-bit parallel samples from the AD9226 synchronous to the conversion clock CLK, buffering one complete A-scan line (typically 2048 to 4096 points) into on-chip block RAM.
- **2. DC Removal (Moving Average Subtractor):** The ADC input signal often contains a slight DC baseline offset caused by amplifier common-mode bias or converter drift. A moving average subtractor computes the local mean across a running window and subtracts it from each incoming sample, centering the RF waveform at zero.
- **3. 5 MHz Digital Bandpass Filter:** The signal passes through a digital Finite Impulse Response (FIR) bandpass filter centered at 5.0 MHz with a 2 MHz passband. This filter rejects low-frequency motor vibrations, 50 Hz power hum and high-frequency quantization noise outside the transducer bandwidth.
- **4. Quadrature Demodulation:** The filtered RF signal is separated into In-Phase (I) and Quadrature (Q) baseband components using digital multiplication and low-pass filtering.

In the quadrature demodulator, the incoming RF signal is split into two parallel channels:

- **In-Phase Channel (I):** Multiplied by a digital cosine wave cos(2*pi*f0*t) generated by an FPGA direct digital synthesizer at local oscillator frequency f0 = 5 MHz. The product contains a baseband component and a high-frequency sum component at 2*f0 (10 MHz). A digital Low Pass Filter (LPF) removes the 10 MHz component, leaving baseband signal I(t).
- **Quadrature Channel (Q):** Multiplied by a 90-degree phase-shifted sine wave -sin(2*pi*f0*t). After matching low-pass filtering, this yields the orthogonal baseband component Q(t).

5. Envelope Detection: The instantaneous amplitude of the acoustic reflection is calculated from the baseband quadrature pairs according to:

$$A(t) = \text{sqrt}(I(t)^2 + Q(t)^2)$$

This removes the 5 MHz carrier oscillation entirely, leaving a smooth positive envelope representing reflection strength at depth t.

- **6. Log Compression:** Because raw ultrasound echo amplitudes span a dynamic range exceeding 60 dB, directly displaying them on an 8-bit monitor would make weak structural echoes invisible while saturating strong specular reflections. Log compression compresses this wide dynamic range using the logarithmic relation y(t) = 20 * log10(A(t)), mapping acoustic reflections into an 8-bit grayscale intensity (0 to 255).
- **7. Host Telemetry Streaming:** The compressed 8-bit envelope values are buffered in on-chip Block RAM and streamed across the 2,000,000 baud UART link to the host personal computer. Each transmitted A-scan line is packaged with hardware synchronization headers and associated with current Cartesian stage coordinates for real-time cross-sectional rendering and volumetric surface reconstruction.

2.6 Host Visualization Software and Gantry Control Application

To process high-speed acoustic telemetry streamed from the FPGA and provide an intuitive operator interface, a dedicated desktop software application titled ultrasound-app was developed in Java 21 [21] using Apache Maven [22]. The software architecture follows an object-oriented modular design divided into decoupled subsystems: hardware serial communication and packet parsing (io), digital signal conditioning and echo peak detection (dsp), multi-threaded data buffering (model), gantry motion supervision (controller), 2D and 3D visualization rendering (ui), medical color palette generation (colormap) and multi-view snapshot report archiving (storage). The codebase is validated by 35 unit test suites covering mathematical coordinate transformations, buffer index boundaries, serial checksum validation and image rasterization.

Communication between the host PC and the Arty S7 FPGA is established over a high-throughput Universal Asynchronous Receiver-Transmitter (UART) serial connection operating at 2,000,000 baud. The FPGA streams binary telemetry packets consisting of a 4-byte synchronization preamble ([0xFF, 0xAA, 0x55, 0x00]) followed by 1024 baseband envelope amplitude bytes representing a single acoustic A-scan line. In the opposite direction, the host software transmits 16-byte binary command frames ([0xA5, 0x5A, CMD, AXIS, DIR, PARAM1..3, CHKSUM]) to control pulser operational modes, configure Time Gain Compensation curves, jog mechanical axes and initiate automated raster scanning routines.

For two-dimensional acoustic imaging, the application provides an interactive 2D B-scan viewport synchronized with a real-time 1D A-scan waveform oscilloscope widget. The A-scan display features an adjustable near-field ring-down blanking gate (0 to 60 samples) that suppresses large initial excitation artifacts caused by acoustic ringing within the transducer faceplate. A dynamic peak detection algorithm tracks the highest amplitude echo returning from target boundaries, computing and displaying the instantaneous penetration depth in millimeters and reflection magnitude on screen. The B-scan rendering engine supports both a Sweep Pass mode that builds an indexed cross-sectional image as the transducer traverses a linear axis and a Continuous Rolling Waterfall mode for real-time oscilloscope monitoring. Interactive Squelch and Contrast Gain sliders allow operators to dynamically adjust noise floor rejection and logarithmic brightness scaling across live or already recorded frozen scans without requiring hardware re-acquisition.

To achieve true volumetric inspection from two-dimensional raster scans, the application integrates a real-time 2.5D and 3D surface rendering engine that maps spatial gantry coordinates (X, Y) to acoustic echo arrival times (Z). The 3D engine implements smooth Lambertian hill-shading and Blinn-Phong diffuse lighting combined with Painter's algorithm depth sorting, ensuring that polygon surfaces render without overlap artifacts or visual tearing during interactive orbital camera rotations. The software provides three distinct 3D visual styles: a 3D Shaded Surface terrain showing continuous elevation contours, a 3D Wireframe Mesh emphasizing structural gradients and an en-face C-scan projection that computes a Maximum Intensity Projection (MIP) across operator-selected depth gates. A dynamic depth-gate colormapping function automatically rescales the full 256-color palette (selectable between medical Bone, thermal Hot, Inferno and standard Grayscale) across the active depth range, enabling subtle millimeter-scale surface reliefs to stand out clearly against background surfaces.

The application provides complete supervisory control over the mechanical Cartesian scanner. To eliminate mechanical racking and maintain carriage orthogonality along the structural guide rails, the software coordinates dual-Y locked stepper commanding, ensuring that both bottom lead screws receive identical step and directional signals. Operators can manually jog axes with selectable increments from 0.1 mm to 25.0 mm or execute an autonomous serpentine

(boustrophedon) chamber scan with a single click. During autonomous chamber scanning, the host transmits the total scan dimensions and indexing pitch to the FPGA, which executes the entire trajectory in hardware and fires the pulser strictly every 40 microsteps (0.100 mm), completely eliminating acceleration skew and motion blur. Upon scan completion, an automated multi-view snapshot suite generates a comprehensive four-in-one report sheet combining isometric, top-down en-face, front XZ and side YZ views with calibrated colorbars, alongside raw CSV data tables containing all physical coordinate measurements.

2.7 Experimental Results and Performance Evaluation

Experimental validation of the mechanical ultrasound scanning system was conducted using the unified Pulser_V2.0_FreeUni PCB, the Arty S7 FPGA acquisition platform and the motorized Cartesian gantry. To evaluate both high-contrast structural boundary detection and complex geometric shape reconstruction, two distinct benchmarking trials were performed: a metallic coin target scan and a submerged machine screw inspection trial.

In both experimental protocols, targets were positioned at the bottom of the central transparent plastic container and submerged under room-temperature distilled water ($c \approx 1480$ m/s). Water immersion provides excellent acoustic coupling between the active face of the DOBO 5P Φ10 transducer and the specimen, preventing transmission loss across air boundaries. The FPGA pulser FSM commanded pulse-echo cycles at a pulse repetition frequency (PRF) of 100 Hz, while the stepper controller executed an automated serpentine raster trajectory, triggering acoustic transceiver acquisitions every 40 microsteps (0.100 mm) along the fast x-axis.

The first experimental benchmark evaluated acoustic reflection fidelity from a metallic coin specimen. The physical experimental test setup is shown in Fig. 15, depicting the single-element immersion transducer positioned normal to the submerged coin surface within the gantry scanning bay. Due to the pronounced acoustic impedance mismatch between water ($Z \approx 1.5$ MRayl) and metal ($Z > 30$ MRayl), incident ultrasonic waves experience a reflection coefficient exceeding 85 percent, producing strong, clean echoes. Fig. 16 displays the resulting two-dimensional cross-sectional B-scan echogram captured as the transducer translated along a single linear scanline across the coin. The image exhibits a sharp, continuous reflection band

corresponding to the upper metal boundary, demonstrating that the AD8331 analog receiver and Time Gain Compensation circuit effectively suppress baseline noise while preserving sub-millimeter axial resolution.

To reconstruct volumetric data from the metallic specimen, the Cartesian gantry performed a two-dimensional raster grid sweep across a 50 mm by 50 mm scanning area. Fig. 17 presents the en-face C-scan top-down acoustic projection rendered by the Java host application. The circular outer perimeter of the coin is clearly resolved, along with observable variations in acoustic reflection amplitude caused by subtle embossed surface reliefs. Fig. 18 shows the corresponding oblique three-dimensional surface topography rendering produced with Lambertian hill-shading and elevation-coded colormapping, confirming that mechanical translation paired with single-element pulse-echo acquisition achieves millimeter-scale topographic surface mapping without requiring expensive multi-element phased arrays.

The second experimental trial evaluated the scanner's capability for non-destructive inspection of intricate geometries and submerged targets. A metallic machine screw was positioned horizontally at the bottom of the water chamber, as shown in the experimental setup photograph in Fig. 19. Fig. 20 displays the longitudinal 2D B-scan ultrasound slice acquired across the axis of the screw. Distinct echo arrival times clearly differentiate the raised circular screw head from the recessed cylindrical shank, validating the temporal discrimination capability of the 12-bit AD9226 digitizer operating at 65 MSPS.

Raster scanning across the entire submerged fixture yielded the reconstructed en-face C-scan top-down intensity map shown in Fig. 21. The post-processed acoustic image reveals the full morphological silhouette of the specimen, accurately resolving the wider head geometry, the continuous cylindrical body and periodic acoustic scattering patterns caused by the screw threads. Quantitative performance metrics obtained across both experimental trials are summarized in Table III and the complete hardware Bill of Materials is listed in Table IV.

2.8 Team Member Contributions

In accordance with senior project guidelines for collaborative projects, the individual engineering responsibilities and deliverables of the team members are summarized below:

- **Nika Durishvili:** Responsible for the primary engineering implementation across electronics, FPGA firmware, digital signal processing and host software development. Designed the complete transceiver circuit schematics and routed the two-layer Pulser_V2.0_FreeUni PCB layout in Altium Designer, including the high-voltage pulser, T/R switch, analog front end receiver, DAC Time Gain Compensation and multi-rail power distribution network. Implemented all Spartan-7 FPGA logic in HDL on the Digilent Arty S7 development board, encompassing the high-speed AD9226 parallel ADC interface, SPI DAC driver, finite impulse response filtering, quadrature demodulation and serial data framing. Developed the host desktop application in Java for real-time B-scan imaging and coordinated electrical bring-up, oscilloscope diagnostics and acoustic testing. Authored the technical documentation, hardware schematics and report chapters covering the analog circuitry, digital logic, signal processing pipeline and acquisition software.
- **Tengiz Gvelesiani:** Contributed to the mechanical assembly and physical construction of the Cartesian scanning platform. Participated in assembling the structural aluminum extrusion frame, mounting the linear guide rails, dual lead screws, timing belts and 3D-printed gantry brackets and assisted with physical alignment of the carriage over the central water tank during mechanical motion testing.

3. Conclusion and Future Work

This project successfully demonstrated the complete design, construction and validation of an automated, low-cost ultrasound scanning platform that replaces expensive multi-channel phased array hardware with a single-element mechanical translation approach. By pairing a 5 MHz immersion transducer with a custom high-voltage pulser, low-noise analog receiver, FPGA-based quadrature demodulation and a motorized Cartesian gantry, we achieved reliable cross-sectional B-scan imaging of submerged targets at a fraction of commercial costs.

The iterative development from our initial testbench (v1.0) to our unified PCB (v2.0) highlighted critical engineering lessons in high-frequency analog design. We identified parasitic ringing in high-voltage switching loops and eliminated baseline receiver noise by transitioning to dedicated ground planes, direct board-to-board PMOD headers and careful differential routing.

Future improvements planned for this platform include:

- **True 3D Volumetric Rendering:** Expanding the Java visualization software with OpenGL voxel shaders to render interactive 3D volumetric views from acquired C-scan raster data.
- **High-Speed USB/Ethernet Streaming:** Replacing the current UART serial link with FTDI high-speed USB 2.0 or Ethernet to increase frame acquisition speed.
- **Higher Frequency Transducers:** Interfacing 10 MHz and 15 MHz focused immersion probes to achieve higher spatial resolution for fine material flaw inspection.

4. References

[1] T. L. Szabo, Diagnostic Ultrasound Imaging: Inside Out, 2nd ed. Oxford, UK: Academic Press, 2014.

[2] K. K. Shung, Diagnostic Ultrasound: Imaging and Blood Flow Measurements, 2nd ed. Boca Raton, FL, USA: CRC Press, 2015.

[3] J. A. Jensen, 'Field: A program for simulating ultrasound systems,' Medical & Biological Engineering & Computing, vol. 34, no. 1, pp. 351-353, 1996.

[4] DOBO Instruments, 'DOBO 5P Φ10: 5.0 MHz Single-Element Immersion Ultrasonic Transducer Technical Specification,' DOBO Ultrasonic Testing Products, 2020.

[5] Microchip Technology, 'MD1213: High-Speed Dual-MOSFET Driver Datasheet,' Microchip Technology Inc., Chandler, AZ, USA, DS20005713A, 2017.

[6] Microchip Technology, 'TC6320: High-Voltage N- and P-Channel Pair Datasheet,' Microchip Technology Inc., Chandler, AZ, USA, DS20005697A, 2017.

[7] Microchip Technology, 'MD0100: High-Voltage Protection T/R Switch Datasheet,' Microchip Technology Inc., Chandler, AZ, USA, DS20005708A, 2017.

[8] Nexperia, 'BAV99: High-Speed Switching Diodes Datasheet,' Nexperia B.V., Nijmegen, The Netherlands, Rev. 8, 2021.

[9] Analog Devices, 'AD8331: Ultralow Noise, Ultralow Distortion VGA Datasheet,' Analog Devices Inc., Norwood, MA, USA, Rev. G, 2016.

[10] Microchip Technology, 'MCP4821: 12-Bit Voltage Output DAC with Internal VREF Datasheet,' Microchip Technology Inc., Chandler, AZ, USA, DS22244B, 2010.

[11] Analog Devices, 'AD9226: 12-Bit, 65 MSPS Monolithic A/D Converter Datasheet,' Analog Devices Inc., Norwood, MA, USA, Rev. E, 2017.

[12] Digilent Inc., 'Arty S7 Reference Manual: Spartan-7 FPGA Development Board,' Digilent Inc., Pullman, WA, USA, 2021.

[13] AMD Xilinx, 'Spartan-7 FPGAs Data Sheet: DC and AC Switching Characteristics,' AMD Xilinx Inc., San Jose, CA, USA, DS189, 2022.

[14] Texas Instruments, 'LM2940/LM2940C 1-A Low Dropout Regulator Datasheet,' Texas Instruments Inc., Dallas, TX, USA, SNVS770I, 2016.

[15] STMicroelectronics, 'LD1117: Adjustable and Fixed Low Drop Positive Voltage Regulator Datasheet,' STMicroelectronics, Geneva, Switzerland, DocID2572 Rev. 34, 2020.

[16] STMicroelectronics, 'L78: Positive Voltage Regulators Datasheet,' STMicroelectronics, Geneva, Switzerland, DocID2143 Rev. 36, 2021.

[17] XLSEMI, 'XL7015: 0.8A 80V High Voltage Step-Down DC-DC Converter Technical Datasheet,' Shanghai XLSEMI Electronic Co., Ltd., Shanghai, China, 2018.

[18] Songle Relay, 'SRD Series Miniature High-Power Electromechanical Relay Datasheet,' Ningbo Songle Relay Co., Ltd., Ningbo, China, SRD-05VDC-SL-C, 2019.

[19] Diodes Incorporated, 'B170/B180/B190/B1100: 1.0A Surface Mount Schottky Barrier Rectifier Datasheet,' Diodes Inc., Plano, TX, USA, DS30018 Rev. 14-2, 2018.

[20] Allegro MicroSystems, 'A4988: DMOS Microstepping Driver with Translator and Overcurrent Protection Datasheet,' Allegro MicroSystems LLC, Manchester, NH, USA, 2020.

[21] Oracle Corporation, 'Java Platform, Standard Edition 21 (Java SE 21) Language and API Specification,' Oracle Corp., Austin, TX, USA, 2023.

[22] Apache Software Foundation, 'Apache Maven Project Management and Build Automation Framework,' Apache Software Foundation, Forest Hill, MD, USA, 2023.

[23] Altium LLC, 'Altium Designer User Guide: Schematic Capture and Multi-Layer PCB Layout,' Altium LLC, San Diego, CA, USA, 2023.

[24] AMD Xilinx, 'Vivado Design Suite User Guide: Logic Simulation and Synthesis,' AMD Xilinx Inc., San Jose, CA, USA, UG900 (v2022.2), 2022.

Appendices

Appendix: Project Source Code and Repository Links

The complete design files, hardware schematics, FPGA firmware RTL, host visualization software and documentation for the single-element mechanical ultrasound scanner are publicly available in the project GitHub repository:

- **Primary GitHub Repository:**

  https://github.com/TengizGvelesiani/ultrasound-device/tree/main

- **Hardware and PCB Directory (PCB/):** Complete Altium Designer project files, multi-sheet schematics, two-layer PCB layout (Pulser_V2.0_FreeUni), Gerber fabrication outputs and Bill of Materials (BOM) for the unified analog front end and high-voltage pulser board.
- **FPGA Firmware Directory (FPGA/):** Xilinx Vivado project files (FPGA.xpr) and modular VHDL RTL source code for the Digilent Arty S7 board, including pulser_fsm.vhd (high-voltage excitation and active damping), tgc_controller.vhd (SPI DAC driver), dsp_pipeline.vhd (quadrature demodulation, FIR low-pass filtering, CORDIC magnitude engine and log_lut.coe compression), acquisition_ram.vhd (1024-byte BRAM buffer and 2 Mbaud UART streamer), uart_rx_protocol.vhd (16-byte host command decoder with XOR checksum validation) and stepper_controller.vhd (synchronized dual-Y and X stepper motion engine with position-triggered acoustic firing every 40 microsteps).
- **Host Software Directory (ultrasound-app/):** Object-oriented Java 21 desktop acquisition and visualization software built with Apache Maven (pom.xml, mvnw.cmd). Contains Swing user interface components, real-time A-scan oscilloscope with ring-down blanking, 2D B-scan sweep and continuous waterfall modes, 2.5D / 3D volumetric surface renderer with Lambertian hill-shading, Blinn-Phong lighting, Painter's algorithm depth sorting, dynamic depth-gate colormapping (Bone, Hot, Inferno and Grayscale), C-scan Maximum Intensity Projection (MIP), interactive 6-DOF orbital camera controls, automated multi-view snapshot report archiving and 35 JUnit 5 test suites.